

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1

Subject: 392133 Physics III

Date: 1 October 2018

Year: 2018

Section: 15-18

Time: 13.00-15.00

Name: _____ ID: _____ Class: _____

Instructions:

1. The examination has 7 pages (including this page), 6 problems and a total score of 60 points.
2. Write all your solution and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are NOT allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

Cheating in the exam is considered an extremely serious
offence which will result in expulsion from the University.

1.1 [5 points] 2 kg block oscillates on the end of the spring with a spring constant of 200 N/m. If the system has an energy of 4.0 J, find the maximum speed of the block.

1.2 [5 points] An ore car of mass 4,000 kg starts from rest and rolls downhill on tracks from a mine. A spring with $k = 400,000$ N/m is located at the end of the tracks. At the spring's maximum compression, the car is at an elevation 10 m lower than its elevation at the starting point. How much is the spring compressed in stopping the ore car? Ignore friction.

2.1 [5 points] A body oscillates with simple harmonic motion along the x axis. Its displacement varies with time according to the equation $x = 5.0 \cos(\pi t)$. Find the magnitude of the acceleration (in m/s^2) of the body at $t = 1.0$ s.

2.2 [5 points] Mr. Peter says that if the length of a simple pendulum is doubled, its period will increase by a factor of 2. Do you agree with him? Please explain.

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3.1 [5 points] A piano string of density 0.0050 kg/m is under a tension of 200 N . Find the velocity with which a wave travels on the string.

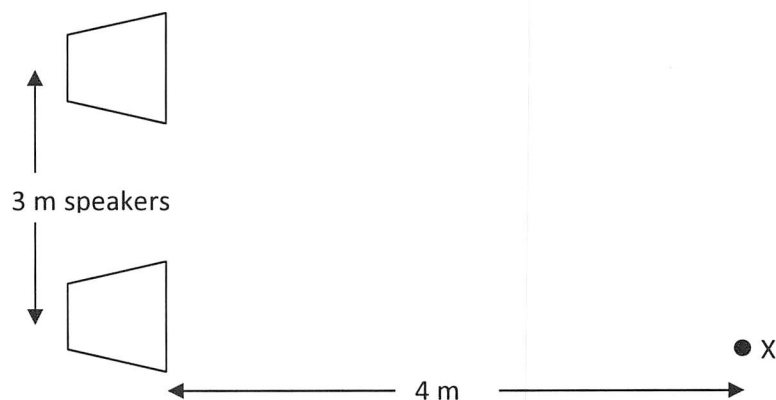
3.2 [5 points] A fire whistle emits a tone of 170 Hz . Take the speed of sound in air to be 340 m/s . Find the wavelength of this sound.

4. [10 points] When a certain string is clamped at both ends, the lowest four resonant frequencies are 50, 100, 150, and 200 Hz. When the string is also clamped at its midpoint, find the lowest four resonant frequencies.

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5.1 [5 points] A point source emits sound waves with a power output of 100 W. What is the sound intensity (in W/m^2) at a distance of 10 m?

5.2 [5 points] Two small identical speakers are connected (in phase) to the same source. The speakers are 3 m apart and at ear level. An observer stands at X, 4 m in front of one speaker as shown. If the amplitudes are not changed, find the wavelength of the sound that he hears at the least intense.



6.1 [5 points] A column of argon is open at one end and closed at the other. The shortest length of such a column that will resonate with a 200 Hz tuning fork is 42.5 cm. Determine the speed of sound in argon.

6.2 [5 points] A truck moving at 3 m/s passes a police car moving at 57 m/s in the opposite direction. If the frequency of the siren is 500 Hz relative to the police car, what is the frequency heard by an observer in the truck after the police car passes the truck? (The speed of sound in air is 343 m/s.)

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